Detailed Project Description: Ultrasonic Distance Measurement System Using Arduino

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Introduction

This project demonstrates a **distance measurement system** using an **Arduino Uno**, an **HC-SR04 ultrasonic sensor**, and a **16×2 I2C LCD display**. The system measures the distance between the sensor and an object by transmitting ultrasonic sound waves and calculating the time taken for the reflected echo to return. The measured distance is displayed in real time on the LCD.

The use of an **I2C-based LCD** reduces the number of required connections, making the circuit simpler and more efficient compared to parallel LCD interfaces.

System Components and Their Roles

Component List

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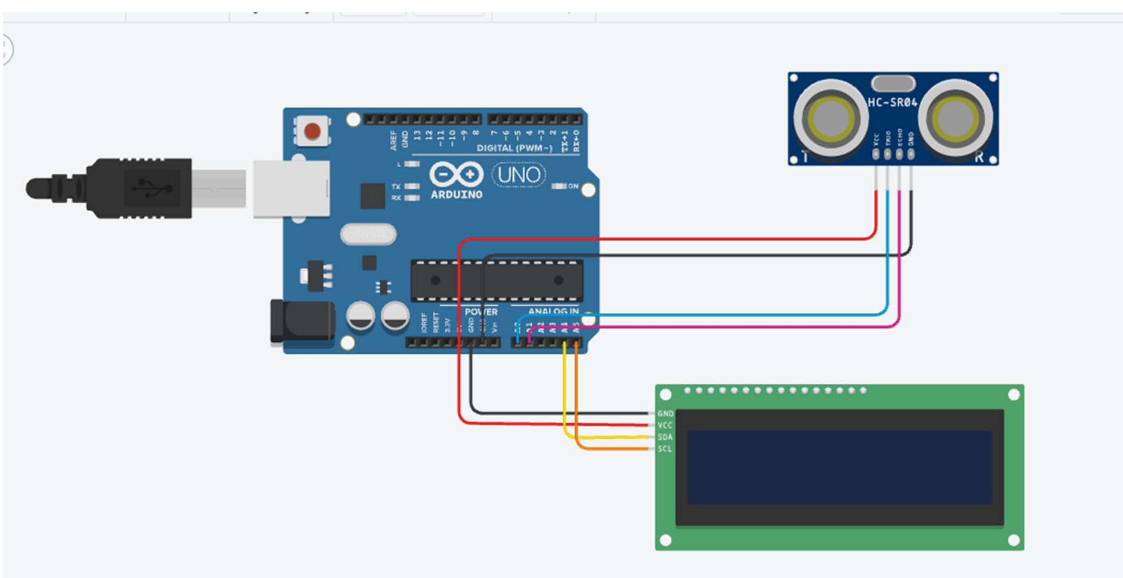
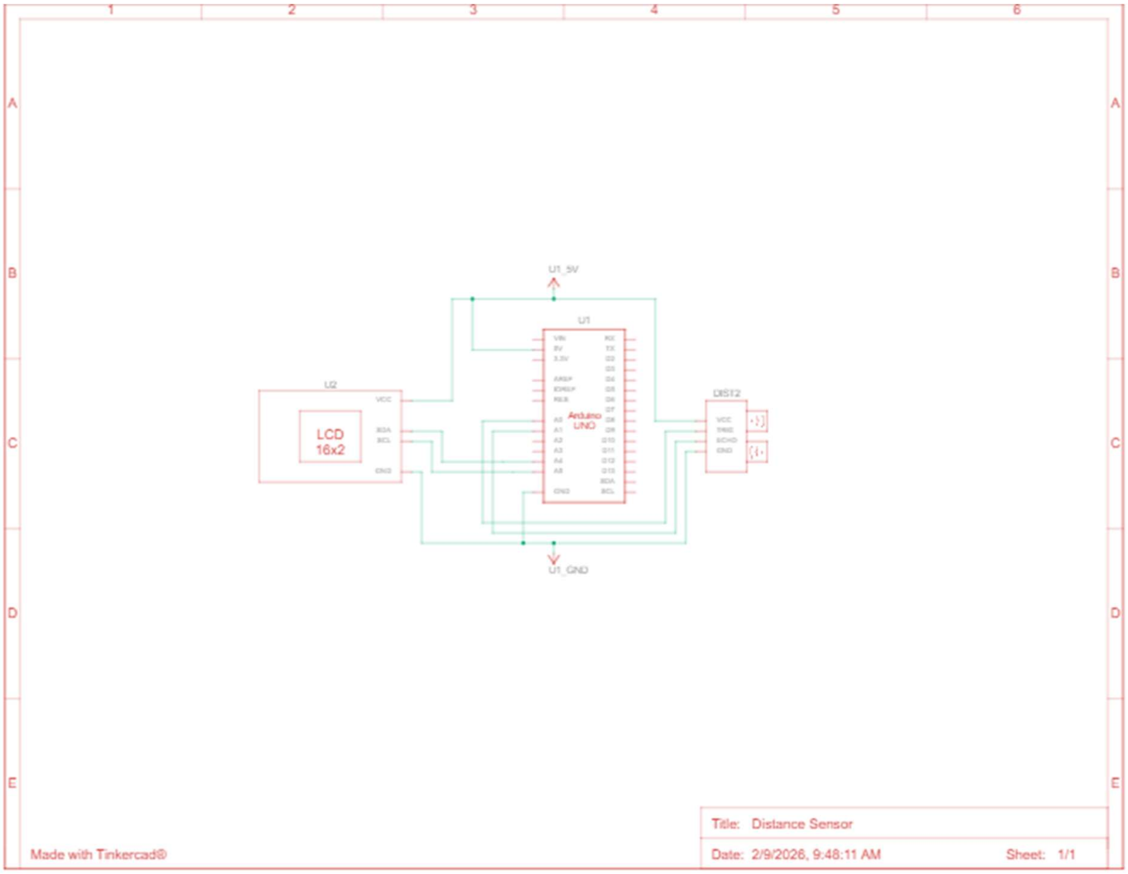
Name	Quantity	Component
U1	1	Arduino Uno R3
U2	1	PCF8574-based, 39 (0x27) LCD 16 x 2 (I2C)
DIST2	1	Ultrasonic Distance Sensor (4-pin)





1. Arduino Uno R3

The Arduino Uno is the **central processing unit** of the system. It:

- Generates trigger pulses for the ultrasonic sensor
- Measures the echo pulse duration
- Calculates distance using timing calculations
- Sends the processed data to the LCD display

The Arduino is powered via USB and distributes 5V power to both the sensor and the LCD.

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2. HC-SR04 Ultrasonic Distance Sensor

The HC-SR04 measures distance using **ultrasonic sound waves (40 kHz)**.

Pin Functions:

- **VCC:** Supplies 5V power
- **GND:** Ground connection
- **TRIG:** Receives a short pulse from the Arduino to start measurement
- **ECHO:** Outputs a pulse whose duration corresponds to the distnce

Working Principle:

1. Arduino sends a 10 µs pulse to the TRIG pin.
2. The sensor emits ultrasonic waves.
3. Waves hit an object and reflect back to the sensor.
4. The ECHO pin stays HIGH for the time taken by the sound to travel to the object and return.
5. The Arduino measures this time and converts it into distance.

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3. 16×2 I2C LCD Display (PCF8574 – Address 0x27)

The LCD is used to **display the measured distance** in centimeters or inches.

Advantages of I2C LCD:

- Requires only two data lines (SDA and SCL)
- Reduces wiring complexity
- Improves readability and circuit organization

Connections:

- **SDA → Arduino A4**
- **SCL → Arduino A5**

The PCF8574 module acts as an I/O expander, allowing the LCD to communicate with the Arduino using the I2C protocol.

Circuit Operation

1. When powered ON, the Arduino initializes the LCD and ultrasonic sensor.
2. The Arduino continuously sends trigger pulses to the HC-SR04.
3. Echo duration is captured using Arduino timing functions.
4. Distance is calculated using the formula:

$$\text{Distance (cm)} = \frac{Echo\ Time\ (\mu s) \times 0.034}{2}$$

5. The calculated distance is displayed on the LCD.
6. The process repeats continuously, updating the display in real time.

CODE:

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

#define trigPin A0 //Sensor Trig pin connected to Arduino pin A0

#define echoPin A1 //Sensor Echo pin connected to Arduino pin A1

long distanceInch;


void setup()

{

 pinMode(trigPin, OUTPUT);

pinMode(echoPin, INPUT);

  lcd.init();

  lcd.backlight();

  lcd.clear();

  lcd.setCursor(0,0);

  lcd.print("Simple  Circuits");

  delay(2000);

  lcd.clear();

  lcd.setCursor(0,0);  //Set LCD cursor to upper left corner, column 0, row 0

  lcd.print("Distance:");//Print Message on First Row

  lcd.setCursor(0,1);

  lcd.print("Distance:");

}
```

```
void loop()
{
  long duration, distance;
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = (duration/2) / 29.1;
  distanceInch = duration*0.0133/2;

  lcd.setCursor(9,0);
  lcd.print("                ");
  lcd.setCursor(9,0);
  lcd.print(          distance); //Print measured distance
  lcd.print(" cm");  //Print your units.
  lcd.setCursor(9,1);  //
  lcd.print("                "); //Print blanks to clear the row
  lcd.setCursor(9,1);
  lcd.print(distanceInch);
  lcd.print(" inch");  //Print your units.
delay(200); //pause to let things settle
}
```

Schematic and Simulation

- The schematic diagram shows correct power distribution, grounding, and signal connections between the Arduino, LCD, and ultrasonic sensor.
- The Tinkercad simulation validates the circuit behavior before physical implementation.

Advantages of the System

- Non-contact distance measurement
- High accuracy for short-range distances
- Simple and low-cost design
- Easy to expand with alarms, buzzers, or wireless modules

Applications

- Obstacle detection systems
- Smart parking assistance
- Water level monitoring
- Robotics navigation
- Security and automation systems

Conclusion

This project successfully implements an **ultrasonic distance measurement system** using Arduino. By integrating an HC-SR04 sensor with an I2C LCD, the system provides accurate real-time distance readings with minimal wiring and efficient communication. The design is flexible and can be easily extended for advanced embedded system applications.